

VO HOA THUAN

Full Stack • Platform Engineering • AI-Integrated Systems

Ho Chi Minh City, Vietnam | (+84) 867 962 672 | vohoathuan.devt@gmail.com

GitHub: github.com/Vo-Hoa-Thuan | LinkedIn: <https://www.linkedin.com/in/thuan-dev2004/> | Profile: thuan.dev

SUMMARY

Full-Stack Engineer with hands-on experience building modern web applications, backend services, and Linux-based infrastructure. Developed AI-integrated systems using OpenAI, Gemini, and Claude APIs while designing scalable backend architectures with Node.js, NestJS, FastAPI, PostgreSQL, and Docker.

Experienced across the full software lifecycle, from React frontend development and API design to deployment-ready environments, server administration, mail infrastructure, and production incident response. Interested in building reliable software platforms that combine strong engineering fundamentals with practical AI capabilities.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL, PHP

Frontend Engineering: React.js, Next.js, Tailwind CSS, Zustand, TanStack Query, Responsive UI Development

Backend Engineering & APIs: Node.js, NestJS (modular architecture, dependency injection, guards/interceptors), Express.js, FastAPI, PHP, RESTful API design, JWT/OAuth2 authentication, service-to-service communication

Databases: PostgreSQL, MySQL, MongoDB, Redis, Prisma ORM — relational schema design, migrations, indexing strategies, query optimization, data modeling

Infrastructure: Linux (Ubuntu/CentOS) administration, Docker & Docker Compose, Nginx reverse proxy, GitHub Actions, shell scripting, multi-service environments, deployment-ready architectures

Mail & Security: Postfix relay configuration, SPF, DKIM, DMARC, SMTP troubleshooting, mail queue diagnostics, Linux hardening, log-based incident investigation, access-control management

AI Integration: OpenAI, Gemini, and Claude APIs, prompt engineering, LLM-assisted workflows, AI-powered content processing, inference service integration, automated explanation generation

PROFESSIONAL EXPERIENCE

System Administrator & Backend Developer | Sieu Toc Viet | Ho Chi Minh City, Vietnam (1- 5/2026)

- AI automation services:** Designed and built 10+ Node.js and Python services that integrate the OpenAI, Gemini, and Claude APIs to automate content generation and data-processing workflows that previously required a team member's daily manual attention. Each service was scoped around a single workflow (e.g. generating or transforming content on a schedule or trigger), making them independently deployable and easy to restart or roll back without affecting the others.
- Linux production administration:** Managed Linux production servers running DirectAdmin and Dockerized services, handling monitoring, patching, and day-to-day operational support.
- Security incident response:** Investigated and remediated malware-related incidents on Linux hosts through process analysis, log review, and system hardening.
- Mail infrastructure operations:** Operated Postfix relay infrastructure, configured SPF/DKIM/DMARC, and resolved SMTP delivery and mail queue issues.
- Infrastructure automation:** Built internal Node.js and Python automation scripts to streamline repetitive operational tasks, data processing, and content publishing workflows across multiple projects.

PROJECTS

ToeicLab VN — AI-Powered TOEIC Learning Platform | Apr 2026 – Personal Project | [GitHub](#)

Stack: NestJS, FastAPI, PostgreSQL, Prisma, Redis, Docker, TypeScript, Python

- **Service split & boundary design:** Split the platform into two independently deployable services instead of one monolith: a NestJS API owning all application state (users, questions, attempts, PostgreSQL via Prisma) and a FastAPI service that does AI inference only. FastAPI holds no database connection — it gets the context it needs over REST from NestJS, calls the LLM, and returns the explanation. This keeps one source of truth for data and lets the inference layer be redeployed or scaled independently.
- **Authentication:** Implemented JWT-based authentication with HttpOnly cookies (rather than storing tokens in localStorage) to reduce exposure to XSS-based token theft, with the NestJS API issuing and validating tokens on protected routes.
- **Data layer:** Designed the relational schema in PostgreSQL using Prisma — modeling users, TOEIC question sets, and attempt/answer history — and managed schema evolution through Prisma migrations rather than hand-written SQL changes.
- **AI explanation workflow:** Built the explanation pipeline so that after a user answers a question, the NestJS API calls the FastAPI service over REST with the question, the user's answer, and the correct answer; FastAPI prompts an LLM to generate a per-question explanation and returns it synchronously to be shown to the user.
- **Infrastructure:** Containerized the NestJS service, FastAPI service, and PostgreSQL with Docker Compose for one-command local setup. Redis is provisioned in the same stack for an upcoming caching layer but isn't yet wired into request handling. Services use environment-based config (no hardcoded hosts) and are deployment-ready, though not yet deployed to a public host.

Postfix Outbound Relay — Mail Infrastructure & DevOps | Jan – Mar 2026 - Personal Project | [GitHub](#)

Stack: Linux, Docker, Postfix, SMTP, SPF, DKIM, DMARC

- **Mail flow & access control:** Built a self-hosted outbound SMTP relay for transactional email: application services authenticate to the relay (rather than sending mail directly), and the relay is locked down so only authenticated, known services can submit mail through it — reducing the risk of the relay being used as an open relay or for spam.
- **Sender authentication:** Configured SPF (authorized sending IPs), DKIM (cryptographic message signing), and DMARC (policy for handling messages that fail SPF/DKIM) on the sending domain to improve sender reputation and reduce inbox-placement failures at receiving mail providers.
- **Deployment:** Containerized the relay with Docker for repeatable, version-controlled deployment — the same configuration can be rebuilt and redeployed on a new host without manual reconfiguration.

HCM_METRO — Metro Management & Tracking Platform | Sep 2025 - Academic Project | [GitHub](#)

Stack: React.js, Node.js, Express.js, MongoDB, TypeScript, Docker

- **Backend architecture:** Built the backend in Node.js/Express.js to handle route management and a tracking simulation, exposing REST endpoints consumed by the React frontend.
- **Data modeling:** Modeled routes, stations, and tracking data as MongoDB collections, with indexes chosen to match the application's main read patterns (e.g. looking up a route's stations and current tracking state) rather than indexing everything by default.
- **Authentication:** Implemented Google OAuth for sign-in and an OTP-based verification step for new sign-ups, covering both third-party and direct registration flows.
- **Deployment:** Containerized with Docker for consistent local development and setup across environments.

EDUCATION

Bachelor of Information Technology 2022 – 2026 (Expected)

Academy of Cryptography Techniques (ACT) | English: technical working proficiency

INTERESTS: AI & Software Engineering | Linux & Platform Infrastructure | Football & Badminton